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20th of February 2026

Material Selection Tool for Marine Piping Systems

Abstract

This project presents a modular material-selection system for piping applications in marine environments, designed to evaluate the compatibility between different materials and a wide range of operating fluids. The aim is to provide a reliable tool that automates decision-making based on technical criteria, reducing errors and standardising the selection process. The system integrates a structured database of materials and fluids, together with a set of general rules—temperature limits, corrosion behaviour—and specific rules defined for each fluid, including steam, CO₂, cryogenic fluids and various oils. By combining both general and fluid-specific rules, the tool identifies suitable materials and generates clear justifications for rejected options. The results demonstrate that the modular approach enables precise, transparent and easily expandable evaluations, making the tool an effective support for engineers in design environments and marine operations.

1. Introduction

Piping systems are key components in marine applications and require adequate material selection to guarantee the safety, reliability, and durability of installations. The wide variety of fluids present in ships and oil rigs — seawater, steam, CO₂, oils, and cryogenic fluids— implies very diverse operative conditions that directly influence material behaviour. Incorrect material selection can cause accelerated corrosion, structural failures or containment losses, with operational and economic consequences.

Traditionally, material selection has been based on engineering experience, regulatory standards, and dispersed criteria. This process can be slow, prone to errors, and difficult to standardise, especially when multiple factors intervene, such as the temperature, corrosivity of the fluid, and brittleness of the material at low temperatures.

2. Objectives

The objective of this project is to develop a modular tool capable of automating the selection of materials for pipes based on fluid properties and material features. This tool aims to reduce the variability in decision making, improve the traceability of applied criteria, and provide reliable support during design and operational stages.

The scope of the project highlights the technical compatibility evaluation without considering economic, logistical or regulatory factors, as the tool is designed as a support for engineers.

3. Methodology

3.1 Overview

The methodology follows a modular, rule-based approach designed to ensure clarity, scalability and traceability in the material-selection process.

3.2 Data Structure: Materials and Fluids

A structured database was established to consolidate the materials typically employed in marine and oil and gas piping applications—namely carbon steel, stainless steel, copper-nickel, GRP, PVC/PE, galvanised steel and copper. For each material, the database records critical engineering properties, including maximum operating temperature, pressure limits, corrosion resistance classification, material category, chemical compatibility considerations and susceptibility to low-temperature embrittlement. These attributes provide the quantitative basis for the rule-driven compatibility evaluation implemented in the system.

A second database was developed to store the fluids considered in the system—sea water, fresh water, ballast water, hydraulic oil, fuel oil, compressed air, CO₂, cryogenic fluids, lubricating oil and steam. For each fluid, the database records the key operational and chemical properties required for compatibility assessment, including minimum and maximum operating temperature, chemical nature, corrosion level and the typical systems in which the fluid is used. These parameters allow the tool to evaluate how each fluid interacts with different materials under realistic marine and industrial conditions.

3.3 System Architecture and Rule-Based Framework

The system is based on a modular architecture that integrates a structured database of materials and fluids, together with a set of general rules — such as temperature limits and corrosion resistance — and specific rules defined for each fluid. By combining both general and fluid-specific rules, the tool determines the compatibility between materials and fluids, identifying suitable options and providing clear justifications for rejected materials.

3.4 General Rules Engine

Temperature limits, corrosion behaviour and chemical compatibility are assessed by cross-referencing the properties defined in the materials and fluids databases. The engine verifies that the operating temperature falls within the material's allowable range, evaluates corrosion resistance against the fluid's corrosion classification and identifies potential incompatibilities related to the chemical nature of the fluid. These general rules constitute the first layer of evaluation and are applied uniformly to all material–fluid pairs.

3.5 Fluid Specific Rules

TABLE 1. SELECTED FLUIDS AND PROPERTIES

Fluid	Specific Rules
Sea water	- Materials with low corrosion resistance are rejected. - Non-metallic materials may be rejected.
Fresh water	Materials with low corrosion resistance are rejected only if the fluid is classified as medium corrosivity .
Ballast water	Same as fresh water and sea water in terms of corrosivity .
Hydraulic oil	Plastic (PVC/PE) and GRP are rejected .
Fuel oil	Plastic (PVC/PE) and GRP are rejected.
Lubricating oil	Only metallic materials are allowed.
Compressed air	Only metallic materials are allowed.
CO ₂	- At normal temperatures, only metallic materials are allowed. - At temperatures below -20°C, only carbon steel and galvanised steel are allowed.
Cryogenic Fluids	- Rejected materials: Plastic (PVC/PE), GRP, copper, and materials with low resistance to extreme temperatures. - Allowed materials: Stainless steel.
Steam	- Only metallic materials allowed. - Materials whose maximum temperature resistance is below 400 °C are rejected.

4. Procedure

1. The user introduces the name of the fluid and its operating temperature and pressure.
2. The tool applies general rules.
3. The tool applies specific rules.
4. The results found in general and specific rules are combined.
5. The tool shows the final classification and justification of rejected materials.

5. Results

5.1 Sea water at 25°C and 100 bar

TABLE 2. RESULTS FOR SEA WATER ANALYSIS

MATERIAL	RESULT	EXPLANATION
Stainless steel	Accepted	Within temperature and pressure ranges and resistance to marine corrosion.
Copper-nickel (90/10)	Rejected	Pressure exceeds material limit (30 bar).
Carbon steel	Rejected	- Pressure exceeds material limit (40 bar). - Material not resistant to high corrosivity. - Chemical incompatibility. - Insufficient corrosion resistance for seawater.
GRP	Rejected	- Pressure exceeds material limit (16 bar). - Material not resistant to high corrosivity. - Chemical incompatibility. - Insufficient corrosion resistance for sea water.
Plastic (PE/PVC)	Rejected	- Pressure exceeds material limit (10 bar). - Material not resistant to high corrosivity. - Chemical incompatibility. - Insufficient corrosion resistance for sea water.
Galvanised steel	Rejected	- Pressure exceeds material limit (25 bar). - Material not resistant to high corrosivity. - Chemical incompatibility. - Insufficient corrosion resistance for sea water.
Copper	Rejected	- Pressure exceeds material limit (20 bar). - Material not resistant to high corrosivity. - Chemical incompatibility. - Insufficient corrosion resistance for sea water.

Stainless steel is the only suitable material for sea water at 25°C and 100 bar. Its corrosion resistance and pressure rating place this material within a safe operating range.

Although copper-nickel (90/10) exhibits excellent corrosion resistance, it is discarded due to the allowed maximum pressure (30 bar), which is lower than the required 100 bar.

The remaining materials — carbon steel, GRP, PE/PVC, galvanised steel, and copper — were rejected due to the combination of insufficient corrosion resistance and maximum pressure lower than operational conditions.

5.2 Fresh water at 60°C and 10 bar

TABLE 3. RESULTS FOR FRESH WATER ANALYSIS

MATERIAL	RESULT	EXPLANATION
Stainless steel	Accepted	Excellent corrosion resistance and wide thermal range.
Copper-nickel (90/10)	Accepted	Chemical compatibility with fresh water.
Carbon steel	Accepted	Allowed pressure and temperature. Good compatibility with fresh water.
GRP	Accepted	Allowed pressure and temperature. Suitable for water systems.
Plastic (PE/PVC)	Accepted	Temperature and pressure within allowable limits.
Galvanised steel	Accepted	Compatible with fresh water; conditions within the limits.
Copper	Accepted	Suitable for fresh water; good thermal resistance.

In this scenario, all materials are suitable for operation with fresh water in these conditions. Temperature and pressure are both within the allowed limits for each material, which also have chemical compatibility with the fluid. Furthermore, fresh water imposes neither corrosion restrictions nor additional requirements, so rejection justifications were not identified.

This case shows a moderate operational scenario, where a wide variety of materials can be used safely and efficiently.

5.3 Hydraulic Oil at 50°C and 20 bar

TABLE 4. RESULTS FOR HYDRAULIC OIL ANALYSIS

MATERIAL	RESULT	EXPLANATION
Stainless steel	Accepted	- Compatibility with hydraulic oils and non-corrosive environments. - Wide thermal range (until 600°C). - High mechanical resistance, suitable for 20 bar. - Excellent durability in closed systems with hydraulic fluids.
Copper-nickel (90/10)	Accepted	- Chemical compatibility with hydraulic oils. - Safe operational temperature (until 300°C). - Allowed pressure (until 30 bar). - Good mechanical behaviour in non-corrosive fluids.
Carbon steel	Accepted	- No risk of chemical wear. - Operational temperature lower than thermal limit (425°C). - Suitable pressure of 20 bar (limit at 40 bar).
GRP	Rejected	- Pressure exceeds material limit (16 bar). - Chemical incompatibility. - Material not suitable for hydraulic oil.
Plastic (PE/PVC)	Rejected	- Pressure exceeds material limit (10 bar). - Chemical incompatibility. - Material not suitable for hydraulic oil.
Galvanised steel	Accepted	- Compatibility with hydraulic oils, without chemical risk. - Adequate pressure (until 25 bar). - Suitable for systems with low corrosion.
Copper	Accepted	- Good chemical compatibility with hydraulic oils. - Safe operational temperature (until 150°C). - Adequate pressure allowance for 20 bar. - Stable material for oil systems at moderate temperature.

The analysis for hydraulic oil at 50°C and 20 bar shows that all metallic materials were evaluated as adequate for these operational conditions. Selected materials have chemical compatibility with hydraulic oils, they maintain mechanical integrity within the pressure and temperature limits, and they do not have additional restrictions in the specific rules of the fluid.

Otherwise, non-metallic materials (GRP and PE/PVC) were rejected due to their incompatibility with oils and pressure limits below operational requirements.

This scenario reflects a moderate environment, where metals supply a reliable and safe performance, whilst polymeric materials and composites are discarded due to limitations inherent to the nature of these materials.

5.4 Cryogenic Fluid at -150°C and 5 bar

TABLE 5. RESULTS FOR CRYOGENIC FLUID ANALYSIS

MATERIAL	RESULT	EXPLANATION
Stainless steel	Accepted	- Suitable for cryogenic applications and non-corrosive environments. - Wide thermal range (until 600°C). - Operating pressure within allowable limits. - Excellent durability in closed systems with cryogenic fluids.
Copper-nickel (90/10)	Rejected	- Chemical incompatibility between fluid and material.
Carbon steel	Rejected	- Chemical incompatibility between fluid and material. - Material becomes brittle at cryogenic temperatures.
GRP	Rejected	- Chemical incompatibility between fluid and material. - Material becomes brittle at cryogenic temperatures. - Cryogenic fluid systems require metallic materials. - Material not suitable for cryogenic fluids.
Plastic (PE/PVC)	Rejected	- Chemical incompatibility between fluid and material. - Material becomes brittle at cryogenic temperatures. - Cryogenic fluid systems require metallic materials. - Material not suitable for cryogenic fluids.
Galvanised steel	Rejected	- Chemical incompatibility between fluid and material. - Material becomes brittle at cryogenic temperatures.
Copper	Rejected	- Chemical incompatibility between fluid and material. - Material becomes brittle at cryogenic temperatures.

Cryogenic fluids require materials with excellent ductility at extremely low temperatures and specific chemical compatibility. Only stainless steel meets these

requirements, maintaining its mechanical integrity and thermal stability in cryogenic conditions.

The rest were discarded due to brittleness at low temperature, chemical incompatibility, or being explicitly rejected in cryogenic systems.

This case study shows a highly restrictive environment, where specialised materials can be used safely.

6. Conclusion

This project successfully developed a material selection tool for marine piping systems that evaluates the compatibility between piping materials and operating fluid conditions. The system is based on the combination of structured databases for both the fluid and material properties to achieve its functionality.

The different case studies demonstrated that the tool can effectively identify both the suitable and rejected materials, while also providing justification for rejected options. These results depend not only on the operational temperature and pressure but also on chemical compatibility, low-temperature performance, and corrosion resistance.

The modular architecture of the system enables the further incorporation of new fluids, materials, conditions, and rules. Further developments could include a larger database, a graphical user interface to improve the usability and visualisation, and the integration of machine-learning techniques to improve the evaluation process and enable the analysis of more complex scenarios. Generally, the project demonstrates how rule-based systems can successfully improve decision-making and consistency in marine piping material selection.